5.2 It is recognized that the applicant may need to provide commercially sensitive information to the Administration in order to demonstrate that a particular Functional Objective has been met by the design of the TWTD and/or would be met in service. In view of this, such information may instead be included in a supporting information annex to the TWTD Technical Report which would not be circulated outside the approving Administration. Where information is provided in that category, it may be cited rather t han being given in full in the TWTD Technical Report itself.

Table 2: Thermal Waste Treatment Device Functional Objectives

These Guidelines have been developed on a technology -neutral basis. Therefore, particular Functional Objectives as listed below may not be applicable to certain types of TWTD since

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SOURCE: MEPC.373(80).pdf
These Guidelines have been developed on a technology -neutral basis. Therefore, particular Functional Objectives as listed below may not be applicable to certain types of TWTD since the point being covered does not exist. The applicant should indicate in the TWTD Technical Report submitted why certain Functional Objectives are not applicable to the TWTD under consideration and provide justification for that assertion.
RESOLUTION MEPC.373(80) (adopted on 7 July 2023)
2023 GUIDELINES FOR THERMAL WASTE TREATMENT DEVICES (TWTD)MEPC 80/17/Add.1
Annex 8, page 9

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Functional objective	Content of TWTD Technical Report
1. Device design and manufacture	
1.1 The device should be designed to meet the Performance Level 1 criteria under all operating conditions with the waste materials it is designed to process	To include those in -service controls and measurements used to regulate the device Description and basis of how the device has been designed and tested to demonstrate the required performance
1.2 Device capacity should be defined	How capacity (i.e. MW, m 3/day, or as applicable) is assessed and defined for the device

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SOURCE: MEPC.373(80).pdf
Description and basis of how the device has been designed and tested to demonstrate the required performance

1.2 Device capacity should be defined How capacity (i.e. MW, m 3/day, or as applicable) is assessed and defined for the device

1.3 The device should be designed so that when installed it will operate as required when the ship is upright and when inclined at any angle of list up to and including 15° either way under static conditions and 22.5° under dynamic conditions (rolling) either way and simultaneously inclined dynamically (pitching) 7.5° by bow or stern Description of how the device has been designed and tested to ensure that it will operate as required under those conditions

1.4 The device should be designed so that there will not be leakage out of the device to the surrounding environment Description and basis of how the device has been so designed and how is that demonstrated and maintained in service

1.5 The device should be designed to handle the various temperatures to which it will be exposed Description and basis of how the device has been so designed and how that is demonstrated and maintained in service

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SOURCE: MEPC.373(80).pdf

maintained in service
1.5 The device should be designed to handle the various temperatures to which it will be exposed
Description and basis of how the device has been so designed and how that is demonstrated and maintained in service

1.6 The device should be designed to resist corrosion and erosion that may be result from the process method applied, the waste materials to be handled or the resulting products Description and basis of how the device has been so designed and how that is demonstrated and maintained in service

1.7 The device should be designed to minimize the amount of by -product, unburnt and partially combusted material in the exhaust gas stream

Description and basis of how the
device has been so designed and
how that is demonstrated and
maintained in service
1.8 The device should be designed to control
intake air flow such that the required
oxygen content and operating conditions
are achieved through the device for it to
function as intended
Description and basis of how the
device has been so designed and
how that is demonstrated and
maintained in service together with
the required oxygen content profile

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SOURCE: MEPC.373(80).pdf
are achieved through the device for it to
function as intended
Description and basis of how the
device has been so designed and
how that is demonstrated and
maintained in service together with
the required oxygen content profile
across the device in operation
1.9 The device should be designed to maintain
the required pressure levels through the
device for it to function as intended
Description and basis of how the
device has been so designed and
how that is demonstrated and
maintained in service together with
the required pressure profile across
the device in operation

Functional objective Content of TWTD Technical Report
1.10 The device should be designed to
minimize visible smoke and particulate
emissions
Description and basis of how the
device has been so designed and
how that is demonstrated and
maintained in service
1.11 The device should be designed to
minimize the formation of dioxins in the
exhaust gas stream when disposing of
garbage containing PVC

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SOURCE: MEPC.373(80).pdf
device has been so designed and how that is demonstrated and maintained in service

1.11 The device should be designed to minimize the formation of dioxins in the exhaust gas stream when disposing of garbage containing PVC

Description and basis of how the device has been so designed and how that is demonstrated and maintained in service

1.12 The device should be designed so that if there is an emergency shutdown, either triggered by the device itself or the user, there will not be abnormal levels of emissions

Description and basis of how the device has been so designed and how that is demonstrated and maintained in service